RFgenerator

Operations Manual
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WARNING
TO REDUCE THE RISK OF FIRE OR ELECTRICAL SHOCK DO NOT EXPOSE THIS EQUIPMENT TO RAIN OR MOSITURE.

WARNING

Only qualified electrical workers should operate and install this system.
Avoid spilling liquids onto/into the unit.
Never operate without earth ground.
Do not expose to excessive heat or moisture.
Do not open – there are no user serviceable parts inside.
Do not block the chassis vent slots or the fan inlet.
Do not operate in an explosive environment.

The RFgenerator is a standalone RF driver designed to generate RF phase and anti-phase outputs. The RFgenerator is designed to drive high capacitive loads. The capacitance of the load is resonated with the RF coil in the RFgenerator. The load capacitance and this coil inductance form a resonate circuit that defines the operating frequency. With high capacitive loads the RF currents can be very high and as a result it can require significant power to generate the desired voltages.

The RFgenerator includes two different communication interfaces, USB, and Ethernet (192.168.1.50 default IP address).

The features of the RFgenerator are listed below:

- Powered with 24VDC supply, up to a maximum of 5 amps
- USB and Ethernet interface with ASCII control commands
- Compatible with MIPS host application
- Load capacitance range, 100pF to 4000pF
- Frequency range of 1MHz to 5MHz
- Phase and anti-phase outputs, 180 degrees phase shift between outputs
- User replaceable coils
- Output voltage range of 5 to 1500Vp-p, this depends on the load
- BNC output connectors, optional MHV for HV outputs
- BNC output monitors
- External DC bias inputs, +/-500V range
- Internal inductance selection relays, user controllable
- Auto tune capability
- RF on indicator
- The system always powers up with the RF disabled
- Control TTL input, supports gating and RF ramp generation
- Optional internal quench module for high-speed RF turn off, 1 cycle
- No power switch, the system powers up when power is applied

This system has the ability to generate high RF voltages and high RF currents. Please be very careful in connecting to your system. If your load is unable to handle the high RF voltage and current you could damage your system and in extreme cases cause a fire.

- Never run the system unattended
- Never apply or remove the load when the system is running
- Never block the cooling holes
- Make sure you use heavy gauge coax to connect to your load, and keep the cables short.

The RFgenerator system is capable of communicating to the host using the USB interface or the Ethernet interface. This Ethernet interface is configured using the host commands defined later in this document. This is a simple interface and does not support name registration or lookup so you need to connect to the IP address assigned to the interface.

The MIPS host application supports connections on port 2015 so if you change the port number the MIPS app will not be able to connect. The MIPS app uses TCP/IP sockets to communicate. You can also make a telnet connection to the RFgenerator Ethernet interface and send any valid RFgenerator host commands.

Front Panel Controls:

24VDC input

Power is applied through a 4 pin DIN power connector. Pin 1 and 2 are +24VDC and Pin 3 and 4 are return. We have provided a 120-watt 24-volt supply, CUI Inc SDI120-24-U-P51 or equivalent. Please use only the supply provided, the DIN power connectors do not have a consistent pin assignment, different models may plug into the RF generator but could cause damage.

Power

This is a green LED that will light when the 24 volt power supply is connected. Note, the system will always power up with the RF output disabled. Even with the outputs disabled the system will generate a few volts of RF.

Interlock

The interlock input controls the RFgenerator RF and DC voltages. If the interlock is open these voltages will not be generated. Note, an eternally supplied bias voltage will not be removed with this interlock system. The interlock signals are available on a Conxall/Switchcraft 4 pin connector, 6282-4PG-3DC (Digikey P/N SC1266-ND), the pin assignment is listed below:

Pin 1 Interlock + input
Pin 2 Interlock - input
Pin 3 Isolated +5V return
Pin 4 Isolated +5V supply

To enable the system, 5 volts must be applied to the Interlock input pins. The interlock input is processed by an opto isolator. The interlock connector provides an isolated 5 volt supply. The interlock can be bypassed by shorting pins 1 to 4 and pins 2 to 3.

To use the interlock system, you can connect pins 2 to 3 and then use a normally closed interlock switch to connect pins 1 to 4. The matting connector is not provided with the system.

Additionally, an internal jumper (JP3), when installed, will bypass the interlock feature. Please refer to your invoice for the configuration of your specific system, note that most systems are shipped with the bypass jumper installed.

USB

The USB interface is designed to communicate with the MIPS host application. The USB interface appears as a virtual com port on your PC and you should use 9600 or 115,200 baud.

Ethernet

The Ethernet interface is provided on a standard RJ45 connector. The default IP address is set to 192.168.1.50 and the port is set to 2015. Host commands allow you to change these parameters as needed. The Ethernet interface commands are the same as the USB interface commands. Both interfaces can be used at the same time.

Control

The Control button has two functions. If you press and release the button quickly the RF output will enable and disable. If you press and hold the button for 3 seconds the system will enter the Auto Tune mode, when the button is release after the 3 second hold. When the system is in Auto Tune mode the status LED will flash. This procedure takes about 1 minute to complete. The RF power level is set at 10% during Auto Tune. Additionally, you can start the Auto Tune process by sending a command from the host computer. The command syntax is defined in the host computer interface section of this document.

Status

The Status orange LED will flash when the system is in Auto Tune mode. This LED will signal additional information in future versions.

RF on

This RF ON red LED is lit with the RF outputs are enabled.

Trig/Gate

This opto isolated control input is not implemented on this version.

Bias

The bias input accepts a DC voltage from -500 volts to +500 volts and applies this DC to both RF outputs. This DC voltage is added to the RF voltage generated by the RFgenerator driver. The RF generator has an internal jumper (JP1) that can be set to use the internal bias supply or an external voltage supplied through this connector. Please refer to your invoice to determine how your system is configured.

Rear Panel Connections:

The rear panel has the RF+ and RF- outputs as well as the monitor outputs. The RF+ and RF- are on BNC connectors.

The monitors are capacitively coupled to the RF outputs using 1500pF Mica capacitors. You can use a scope to monitor the RF on these outputs. Use a 10:1 or 100:1 scope probe. You can install BNC shorting capacitors on these BNC connectors to create a dummy load for the RFgenerator, this will allow you to test the system and provide a capacitive load similar to a SLIM board.

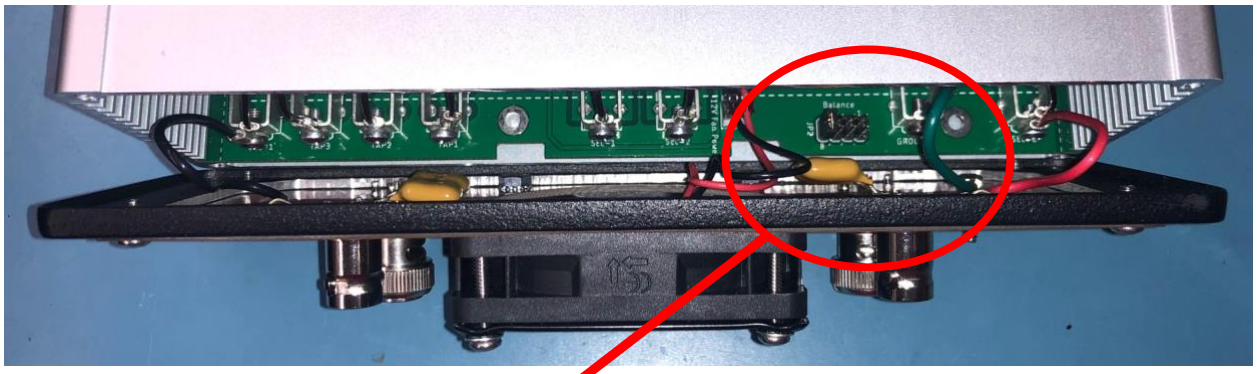
Note! The system is shipped with the sorting caps installed, you can test in this configuration but make sure and remove the shorting caps when testing with your actual load.

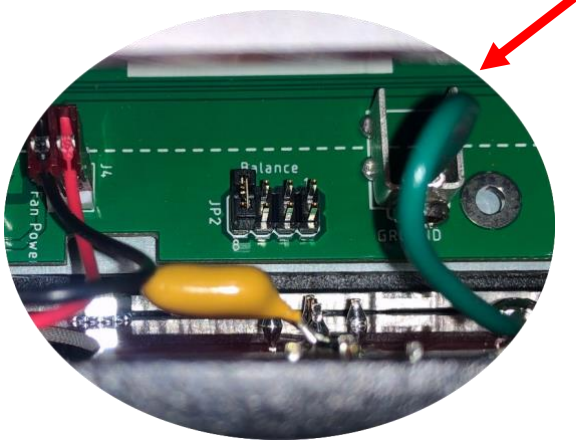
Grounding:

> **You will find a grounding lug on the rear panel. This lug is used to bond the RFGenerator chassis to earth ground. It is critical that this is the first connection made and the last connection removed when working with the RFgenerator.**

Balance Adjustments:

The balance adjustment allows you to adjust the capacitance to ground on the RF+ output to affect the balance. A perfectly balanced system will have the same RF level on the RF+ and RF- outputs. If the RF voltage RF+ and RF- do not match with in your requirements you can use the balance jumpers to make corrections. To access the balance jumpers you will need to remove the rear panel mounting screws and carefully pull the rear panel back to gain access. Details can be seen in the pictures below. These jumpers enable capacitance to be added to the RF+ side of the RFgenerator to ground. The figure below shows the jumpers and lists the capacitance options depending on the number of jumpers installed. Increasing the capacitance will lower the voltage on the RF+ phase and raise it on the RF- phase. Changing the capacitance will also slightly change the resonate frequency. Adjust the jumpers to achieve the desired balance. Make sure the power is off before the jumpers are changed. Also make sure you save the RFgenerator parameters before you turn off the power.





Install jumpers to add capacitance from the RF+ output to ground. The capacitance for each jumper position is listed below. Multiple jumpers can be installed to increase capacitance.

1 = 10pF
2 = 20pF
3 = 47pF
4 = 82pF

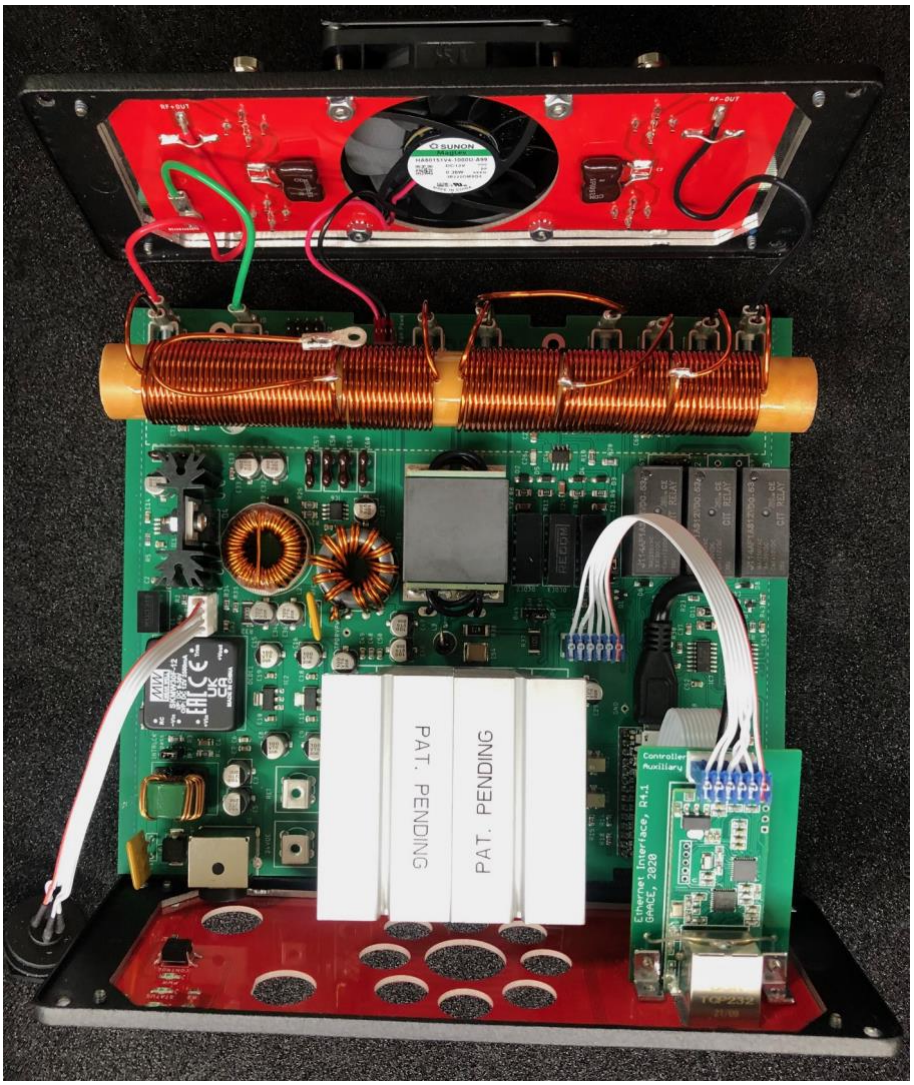
Disassembly:

The RFgenerator has user replaceable coils and internal jumpers. This procedure outlines the steps to disassemble the system.

1.) Remove all connections to the RFgenerator.
2.) Remove the 4 screws that secure the rear panel, shown circled in blue below. You can pull the rear panel back a bit but do not move it more than 0.5”

3.) Remove all the screws shown below circled in blue, also remove the nuts securing the interlock connector. Remove the four screws in the corners of the front panel last. After all the screws and the nut is removed you can pull the front panel forward, in the lower left corner on the inside of the panel you will find a 10 pin IDC connector that you will need to unplug. You can now remove the front panel.

4.) Carefully slide the internal PCB out of the case as you guide the rear panel. Please note the rear panel is connected to the PCB with a number of cables so it will not come free of the system. Below is a picture of the PCB out of the case and reconnected to the front panel for testing.



5.) The Coil can be seen at the top of the picture. The coil form is held in place with standoffs and all coil leads are attached with screw lugs. The coil can be replaced with a few simple hand tools.
6.) Close to the center of the board you will find a jumper, JP1, to select internal or external Bias supply. The system is shipped with this jumper set to use the external Bias supply unless noted otherwise on your invoice. You can set this to Ext to use the front panel Bias SMA connector.
7.) Reverse this procedure to reassemble the RFgenerator.

Coil design

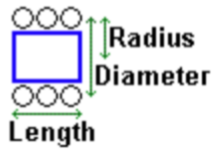
The coil is design for the user's application and can be changed in the field if needed. Internal relays allow you to change the inductance and thus change the resonate frequency.

The resonate frequency is determined by the following equation:

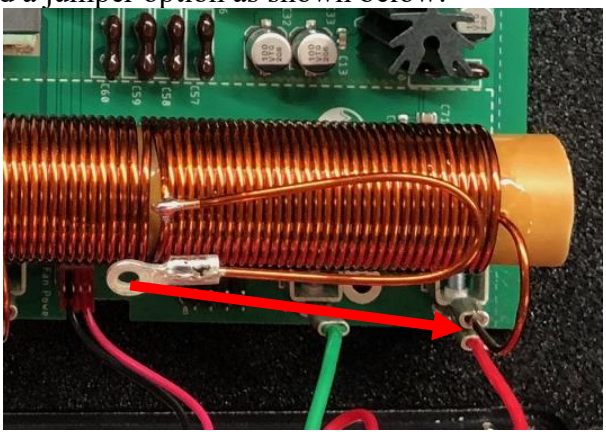
$f = 1 / (2 * \pi * \sqrt{(L * C)})$
Where:
L = inductance is Henries
C = capacitance in Farads

The coil inductance can be calculated using the following equation:
(The following inductance formula requires units in inches.)

$$L = \frac{\text{Radius}^2 \times \text{Turns}^2}{9\,\text{Radius} + 10\,\text{Length}}$$



The default coil design uses a 0.75" coil form and is wound using 16-gauge magnet wire as you can see in the pictures. The three relays are connected in a way to short out the three sections on the far-right side of the coil. The coil details are documented on your invoice. You may find a jumper option as shown below:



You can connect the lug to the terminal screw block, as shown by the arrow, to increase the frequency range of the RF generator. Note, the lug will be protected with heat shrink tubing that you will need to remove first.

Calibration procedure

The RFgenerator has two calibration procedures built into the control firmware, one for the internal Bias supply and a second for the RF output level sensors. These two procedures are outlined below.

DC bias calibration

The internal DC bias supply can be calibrated using the procedure outlined below. This procedure requires the internal Bias jumper to be set for Internal mode and the RF outputs must be disabled. You will need a volt meter to monitor the DC voltage on either the RF+ or RF- output.

The RFgenerator will need to be connected to the MIPS host app and you will use the Terminal interface to issue commands and enter data.

The following steps outline the procedure:

1.) Enter the command *CALBIAS* you will see the following message:
    This function will generate the calibration
    parameters for the DC bias supply. Make sure
    the RF is disabled and measure the DC voltage
    on the RF+ output.
    Press enter to continue...
2.) After you have the voltmeter connected to record the voltage press the enter key. You will see the following message:
    Enter measured voltage:
3.) Enter the value on the voltmeter, is should be around 1.2 volts. After you enter the value press enter and you will see the following message:
    Enter measured voltage:
4.) Enter the value on the voltmeter, this time it is should be around 42 volts. After you enter the value press enter the system will display the calculated calibration parameters and exit the procedure.
5.) At this point the calibration has been generated but not saved, enter the command *SAVE* to store the parameters in nonvolatile memory.

RF level calibration

The RFgenerator uses a pricewise linear lookup table to convert the data from its level detectors into the displayed values. The system is calibrated in the factory but you many want to calibrate it in your specific application. The two output channels are calibrated separately. The system will need to be connected you your load and tuned to your normal operating conditions. You will need an oscilloscope to monitor the output voltages. You can calibrate using the voltage monitors on the rear panel or at any point in your system that you wish to control.

The RFgenerator will need to be connected to the MIPS host app and you will use the Terminal interface to issue commands and enter data. You will need to operate your system over the range you wish to calibrate. In this calibration procedure you will set RF drive levels and then enter voltage readings from your oscilloscope. Before you start the calibration first determine the set of RF drive levels you wish to use. Select a maximum of 8 points and a minimum of 4.

The calibration procedure will be run twice, once for each output phase.

The following steps outline the procedure:

6.) Enter the command *CALRFL,RF+* you will see the following message:

This function will generate a piecewise linear
calibration table. Set the drive level to reach
desired calibration points and then enter the
measured value to the nearest volt. Press enter
When finished. Voltage must be increasing!

Point 1
Set drive level, %:

7.) Enter the RF drive level for this calibration point. Please note, this is a very simple interface and backspace is not supported. To exit this procedure press enter with no other characters and the system will exit and calculate the calibration table with the data entered.

8.) If you entered a drive level the system will set the drive to the value entered and then ask for the measured value, you will see the following message:

Enter measured voltage:

9.) Enter the RF level measured with the oscilloscope at your calibration point. Press enter with no other characters to exit this procedure and calculate the calibration table.

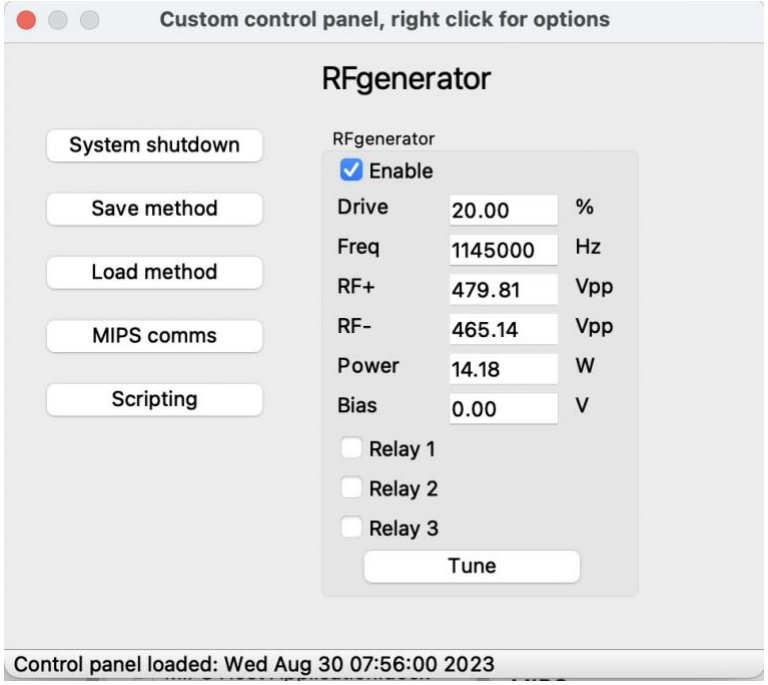
10.) Repeat steps 2 thru 4 until you have finished the calibration for the selected channel.

11.) Use the command *CALRFL,RF-* and repeat the above procedure for the other RF output phase. Make sure you move your oscilloscope probe to the RF- monitoring point.

12.) The calibration data is only in volatile memory when the procedure is completed, enter the command *SAVE* to store the data in non-volatile memory. After entering the *SAVE* command, the RFgenerator will load the calibration table every time the system starts up.

MIPS App Interface

Interfacing with the MIPS host application and using the control panel feature is the best way to use this system and allows monitoring all the critical parameters. Below is a screen capture of the RFgenerators's parameters displayed on a control panel:



The control panel configuration file commands use to generate the RFgenerator dialog are listed below:

```
GroupBox,RFgenerator,200,300,200,50
Ccontrol,Enable,RFgen,CheckBox,SENA_FALSE,SENA_TRUE,GENA_TRUE_FALSE,,10,20
Ccontrol,Drive,RFgen,LineEdit,GDRV,SDRV,,%,10,45
Ccontrol,Freq,RFgen,LineEdit,GFREQ,SFREQ,,Hz,10,70
Ccontrol,RF+,RFgen,LineEdit,,,GRFAVP,Vpp,10,95
Ccontrol,RF-,RFgen,LineEdit,,,GRFAVN,Vpp,10,120
Ccontrol,Power,RFgen,LineEdit,,,GPWR,W,10,145
Ccontrol,Bias,RFgen,LineEdit,GBIAS,SBIAS,,V,10,170
Ccontrol,Relay 1,RFgen,CheckBox,SRLY_1_OPEN,SRLY_1_CLOSE,GRLY_1_CLOSE_OPEN,,10,195
Ccontrol,Relay 2,RFgen,CheckBox,SRLY_2_OPEN,SRLY_2_CLOSE,GRLY_2_CLOSE_OPEN,,10,220
Ccontrol,Relay 3,RFgen,CheckBox,SRLY_3_OPEN,SRLY_3_CLOSE,GRLY_3_CLOSE_OPEN,,10,245
Ccontrol,Tune,RFgen,Button,,TUNE_TRUE,,,20,265
GroupBoxEnd
```

Host Computer Interface:

All commands are ASCII text.
All commands sent to the RFgenerator are terminated with a carriage return (CR) or line feed (LF) character.
All messages sent from the RFgenerator are terminated with a CR and LF.
After a command is received (after the CR or LF is received) RFgenerator will respond with an ASCII ACK (0x06) then a CR or an ASCII NAK (0x15) then a “?” character and a finally a CR. A NAK indicates the command or its arguments was not understood. When a NAK is sent the “?” character is sent to inform users communicating with a terminal emulator that the command was not understood.

Commands:

GVER

Returns the current version string.

GERR

Returns an integer error code indicating the reason for the last NAK. This value is never cleared and always indicates the communications last error.

GNAME

Returns a string that contains the name of the RFgenerator system.

SNAME,Name

This command will set the RFgenerator system name. The Name value is a user defined string. Make sure you issue a Save command to save the changes to flash memory or this name will be lost when power is cycled.

SAVE

This command will save the RFgenerator controller parameters to the flash memory.

RESTORE

This command will read the current settings from the flash memory.

FORMAT

RESET

This command will cause the RFgenerator control computer to reboot.

MUTE, Value

This command will stop all messages from the RFgenerator system to the host computer. The received commands will still be processed but no response will be issued. The value parameter turns this mode on and off where:

Value = OFF to turn the responses on and ON to silence the RFgenerator system.

ECHO,TRUE or FALSE

Setting ECHO true will cause RFgenerator to include the command with all responses.

GCMDS

This command will cause the RFgenerator system to display a list of valid commands that it understands.

THREADS

The RFgenerator system uses a threading control and each module runs on its own Thread as well as a number of threads for control functions. This command displays all the threads and few key parameters. This command is used to monitor the system and check status, mainly a diagnostic tool.

STHRDENA,Thread name,TRUE or FALSE

This command allows you to enable or disable a Thread. This is done by using the thread name and defining it as TRUE or FALSE. If set to FALSE then the thread will stop. This is development and diagnostic tool.

SENA,TRUE or FALSE

This command will enable or disable the RF outputs. TRUE will enable and FALSE will disable.

GENA

This command will return the enable status, TRUE or FALSE.

SFREQ,Freq

This command will set the RF frequency to the value defined my Freq, in Hz.

GFREQ

This command returns the current RF frequency

SDRV,Value

This command defines the drive level for the RFgenerator. The drive level (0 to 100) defines how much power is sent to the output amplifier in the RFgenerator. This drive level affects the RF output, the higher the drive level the higher the output voltage.

GDRV

This command returns the current RF drive level.

SRFV,Value

This command defines the desired (or setpoint) peak to peak voltage on each output of the RFgenerator. This command only applies when the RFgenerator mode has been set to closed loop.

GRFV

This command returns the current peak to peak desired (or setpoint) voltage.

GRFAVP

This command returns the current peak to peak output voltage on the RF+ output.

GRFAVN

This command returns the current peak to peak output voltage on the RF- output.

GDRVV

This command returns the drive voltage sent to the RF amplifier in the RFgenerator.

GDRVI

This command returns the current in amps that is being consumed by the RFgenerator.

GPWR

This command returns the power in watts that is being consumed by the RF amplifier in the RFgenerator. Note, this power is dissipated in the RF amplifier, the coil and some of this power is dissipated in the load.

SGAIN,Value

This command allows you to set the gain of the closed loop control mode. This is a unitless value that defines the closed loop performance.

GGAIN

This command returns the current closed loop gain setting.

SMODE,TRUE or FALSE

This command sets the RFgenerator mode. If TRUE the system is in closed loop mode, in this mode the firmware automatically adjusts the drive level to hold the output voltage at the value defined by the SRFV command. If FALSE the drive level is manually set by the SDRV command.

GMODE

This command returns the current mode.

SMDRV,Value

This command sets the maximum drive level that is allowed. This is a command that allows you to set an upper limit to protect your system.

GMDRV

This command returns the current maximum RF drive limit setting.

SMPWR,Pwr

This command allows you to define a maximum RF power level that is permitted. If the power exceeds this level the firmware will automatically reduce the drive.

GMPWR

This command returns the current maximum power setting.

SRLY,num,OPEN or CLOSE

The RFgenerator has three internal relays that can be used to change the inductance of the internal coil. The inductance is maximum when all three relays are open and minimum when all three are closed. The system will only accept relay commands when the RF is disabled. The num parameters selects the relay, valid selections are 1, 2, or 3. Valid option for the SRLY command are OPEN and CLOSE.

GRLY,num

This command returns the state of the relay defined by num, num selects the relay, 1, 2, or 3. The relay state is returned as OPEN or CLOSE.

SBIAS, voltage

This command will set the internal bias supply to the voltage value entered.

GBIAS

This command returns the internal bias supply's current voltage value.

TUNE,TRUE

This command will start the auto tune process. This is the same procedure that is started using the front panel pushbutton. The system will respond with the following message as soon as the procedure starts:

Tuning…

After the tuning process completes the system will respond with the following message:

Auto tune complete, frequency = xxxxxx

During the auto tune process the drive level is set to 10% and it will remain at 10% when complete. Please note the frequency found will not be saved so make sure and issue the SAVE command to save the frequency to non-volatile memory.

CALBIAS

This command is used to calibrate the internal +/-50V bias supply.

CALRFL,RF+ or RF-

This command is used to calibrate the RF+ and RF- output level monitors. This is a factory setup command and should not be used.

Ethernet commands:

The RFgenerator system automatically detects the Ethernet module on power up and enables the following commands if the module is found. These commands should be set using the USB host interface.

GEIP

This command will return the current IP address used by the Ethernet module.

SEIP,<xxx.xxx.xxx.xxx>

This command allows you to define the Ethernet module's IP address.

GEPORT

This command returns the port number used for telnet connections to the MIPS system using the Ethernet interface.

SEPORT,<Port>

This command sets the port number used for telnet connections to the MIPS system, the default value is 2015 and this is the value expected by the MIPS host computer application.

Where:

Port = TCP/IP port number

GEGATE

This command will return the gateway IP address used by the Ethernet module.

SEGATE,<xxx.xxx.xxx.xxx>

This command allows you to define the gateway IP address.

GESNIP

This command will return the subnet IP address used by the Ethernet module.

SESNIP,<xxx.xxx.xxx.xxx>

This command allows you to define the subnet IP address.

Warrantee

GAA Custom Electronics, LLC warrantees the MIPS system to be free from defects in materials and workmanship and will repair or replace the unit for a period of one year. This warrantee assumes the system is operated in compliance with the procedures and recommendation outlines in this document. GAA Custom Electronics, LLC will also provide free phone support and firmware bug fixes for up to one year. The addition of new features is not covered in this warrantee.

Liability

The liability of GAA Custom Electronics, LLC hereunder or otherwise is solely and exclusively limited to replacement, repair or credit at the purchase price, as GAA Custom Electronics, LLC may elect, for any product which is returned by Buyer during

the applicable warranty period, or services for which timely notice of defect has been given by Buyer, and which are found by GAA Custom Electronics, LLC to be subject to adjustment under this warranty. IN NO EVENT SHALL GAA Custom Electronics, LLC BE LIABLE FOR SPECIAL, INDIRECT, INCIDENTAL OR CONSEQUENTIAL DAMAGES, LOSS OF ANTICIPATED PROFIT OR OTHER ECONOMIC LOSS OR FOR ANY DAMAGES ARISING IN TORT WHETHER BY REASON OF STRICT LIABILITY, NEGLIGENCE OR OTHERWISE.